



Product Backlog

Epic	Aa Name	Status	Description
Documentation	⦿ <u>Requirements</u>	Completed	
Documentation	⦿ <u>Use Case Diagram</u>	Completed	
Documentation	⦿ <u>Use Case Descriptions</u>	Completed	
Documentation	⦿ <u>Activity Diagrams</u>	Completed	
Documentation	⦿ <u>System Sequence Diagrams</u>	Completed	
Documentation	⦿ <u>Test Cases</u>	Completed	
Documentation	⦿ <u>Domain Model</u>	Completed	
Documentation	⦿ <u>Conceptual & Global ER Diagrams</u>	Completed	
Documentation	⦿ <u>Class Diagram</u>	Completed	
Documentation	⦿ <u>Sequence Diagram</u>	Completed	
IdentityService	⦿ <u>Implement domain entities</u>	Completed	User, EmailConfirmationToken, PasswordResetToken
IdentityService	⦿ <u>Create IdentityDbContext</u>	Completed	
IdentityService	⦿ <u>Define IUserService interface</u>	Completed	for user creation, lookup, token generation, credential validation. password reset/change

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IdentityService	◎ <u>Implement UserService</u>	Completed	with EF persistence, token generation, password operations and validation
Testing	◎ <u>Unit tests for UserService</u>	Completed	covering create, duplicate email, validate credentials, change/reset flows
IdentityService	◎ <u>Define IJwtService interface</u>	Completed	for token generation and validation
IdentityService	◎ <u>Implement JwtService</u>	Completed	with configured claims and signing
Testing	◎ <u>Unit tests for JwtService</u>	Completed	assertions on claims and expirations
IdentityService	◎ <u>Define IPasswordHasher interface</u>	Completed	for hashing/verifying passwords
IdentityService	◎ <u>Implement PasswordHasher</u>	Completed	
Testing	◎ <u>Unit tests for PasswordHasher</u>	Completed	validating hash/verify roundtrip and edge cases
IdentityService	◎ <u>Define IEmailSender interface</u>	Completed	for sending emails
IdentityService	◎ <u>Implement EmailSender variants and config binding</u>	Completed	
Testing	◎ <u>Unit tests for EmailSender</u>	Completed	mocking email failures and success behavior
IdentityService	◎ <u>Define IConfirmationLinkGenerator interface</u>	Completed	confirmation/reset link generator abstraction for email content
IdentityService	◎ <u>Implement ConfirmationLinkGenerator</u>	Completed	using request context and config
Testing	◎ <u>Unit tests for ConfirmLinkGenerator</u>	Completed	for link formatting and validity
IdentityService	◎ <u>Implement AuthController</u>	Completed	register, confirm, resend, login, logout, me, change-password,

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	<u>endpoints</u>		reset-password
Testing	⦿ <u>Unit tests for AuthController</u>	Completed	ensuring correct behavior and error mapping
IdentityService	⦿ <u>Wire Dependency Injection in IdentityServiceProgram</u>	Completed	
Testing	⦿ <u>Integration tests for authentication</u>	Completed	for register, login, confirm, resend, forgot, reset flows
Testing	⦿ <u>End-to-end tests for registering, confirming email and logging in</u>	Completed	
LinkingService	⦿ <u>Implement domain entities</u>	Completed	ProviderAccount, TokenSet, ProviderType
LinkingService	⦿ <u>Create LinkingDbContext</u>	Completed	
LinkingService	⦿ <u>Define IOAuthClient interface</u>	Completed	for exchange, refresh, revoke and token parsing
LinkingService	⦿ <u>Implement OAuthClient for Google and Microsoft</u>	Completed	
Testing	⦿ <u>Unit tests for OAuthClient implementations</u>	Completed	ensuring correct exchange, refresh, and revoke behaviors and error handling
LinkingService	⦿ <u>Define IStateStore interface</u>	Completed	for PKCE state storage
LinkingService	⦿ <u>Implement CacheStateStore</u>	Completed	using MemoryCache with TTL/expiry
Testing	⦿ <u>Unit tests for CacheStateStore</u>	Completed	for save/take, expiry and concurrency handling
LinkingService	⦿ <u>Define ITokenStore interface</u>	Completed	for storing encrypted token sets for user/provider
LinkingService	⦿ <u>Implement EfTokenStore</u>	Completed	DataProtection encryption and EF persistence
Testing	⦿ <u>Unit tests for TokenStore</u>	Completed	for encryption roundtrip and persistence edge cases

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LinkingService	⦿ <u>Define IFileProvider and IFileProviderFactory interfaces</u>	Completed	
LinkingService	⦿ <u>Implement FileProviderFactory and providers</u>	Completed	for Google and Microsoft
Testing	⦿ <u>Unit tests for FileProviders</u>	Completed	for factory selection and provider behavior
LinkingService	⦿ <u>Implement LinkProviderService</u>	Completed	start, connect callback, refresh/ensure access, disconnect
Testing	⦿ <u>Unit tests for LinkProviderService</u>	Completed	for start, callback, refresh and revoke using mocks
LinkingService	⦿ <u>Implement ConnectController endpoints</u>	Completed	API endpoints for start, callback, disconnect, status
Testing	⦿ <u>Unit tests for ConnectController</u>	Completed	test responses and redirects
Testing	⦿ <u>Integration tests for ConnectController</u>	Completed	
LinkingService	⦿ <u>Wire Dependency Injection in LinkingServiceProgram</u>	Completed	for IOAuthClient implementations, state store and token store
Testing	⦿ <u>Integration tests for FileProviderFactory and OAuth flows and error handling</u>	Completed	
Testing	⦿ <u>End-to-end test for connecting a provider account</u>	Completed	OAuth start → callback → token persistence workflow
Frontend	⦿ <u>Implement API wrapper functions</u>	Completed	auth.ts and connections.ts
Frontend	⦿ <u>Implement "register" and "login" pages</u>	Completed	

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Testing	⦿ <u>Unit tests for "register" and "login" pages</u>	Completed	
Frontend	⦿ <u>Implement "email confirmed" page</u>	Completed	
Testing	⦿ <u>Unit tests for "email confirmed" page</u>	Completed	
Frontend	⦿ <u>Implement "connections" page</u>	Completed	list providers with connect/disconnect actions and status
Testing	⦿ <u>Unit tests for "connections" page</u>	Completed	
Frontend	⦿ <u>Implement "link result" page</u>	Completed	show link success/failure
Testing	⦿ <u>Unit tests for "link result" page</u>	Completed	
Frontend	⦿ <u>Implement "files" page</u>	Completed	
Testing	⦿ <u>Unit tests for "files" page</u>	Completed	
Frontend	⦿ <u>Implement "account" page</u>	Completed	
Testing	⦿ <u>Unit tests for "account" page</u>	Completed	
Frontend	⦿ <u>Implement "forgot password" and "reset password" pages</u>	Completed	
Testing	⦿ <u>Unit tests for "forgot password" and "reset password" pages</u>	Completed	
Frontend	⦿ <u>Implement "landing" and "not found" pages</u>	Completed	
Testing	⦿ <u>Unit tests for "landing" and "not found" pages</u>	Completed	

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Frontend	⦿ <u>Implement routes</u>	Completed	
Frontend	⦿ <u>Implement reusable components</u>	Completed	
Testing	⦿ <u>Unit tests for reusable components</u>	Completed	
Frontend	⦿ <u>Implement PKCE flow client logic</u>	Completed	follow redirect to provider and handle callback to "link result" page
Testing	⦿ <u>Unit tests for client logic</u>	Completed	
Frontend	⦿ <u>Store JWT in localStorage</u>	Completed	add helper auth utils to attach Authorization header